

GraphRAG Benchmark Report

A 2 million token scientific-paper benchmark comparing LLM-only, Basic RAG, and NetworkX GraphRAG across efficiency, cost, and answer quality.

Dataset

armanc/scientific_papers, arXiv split

Token scale

2,004,563 tokens

Documents

220 papers

Evaluation set

40 grounded questions

Deployment

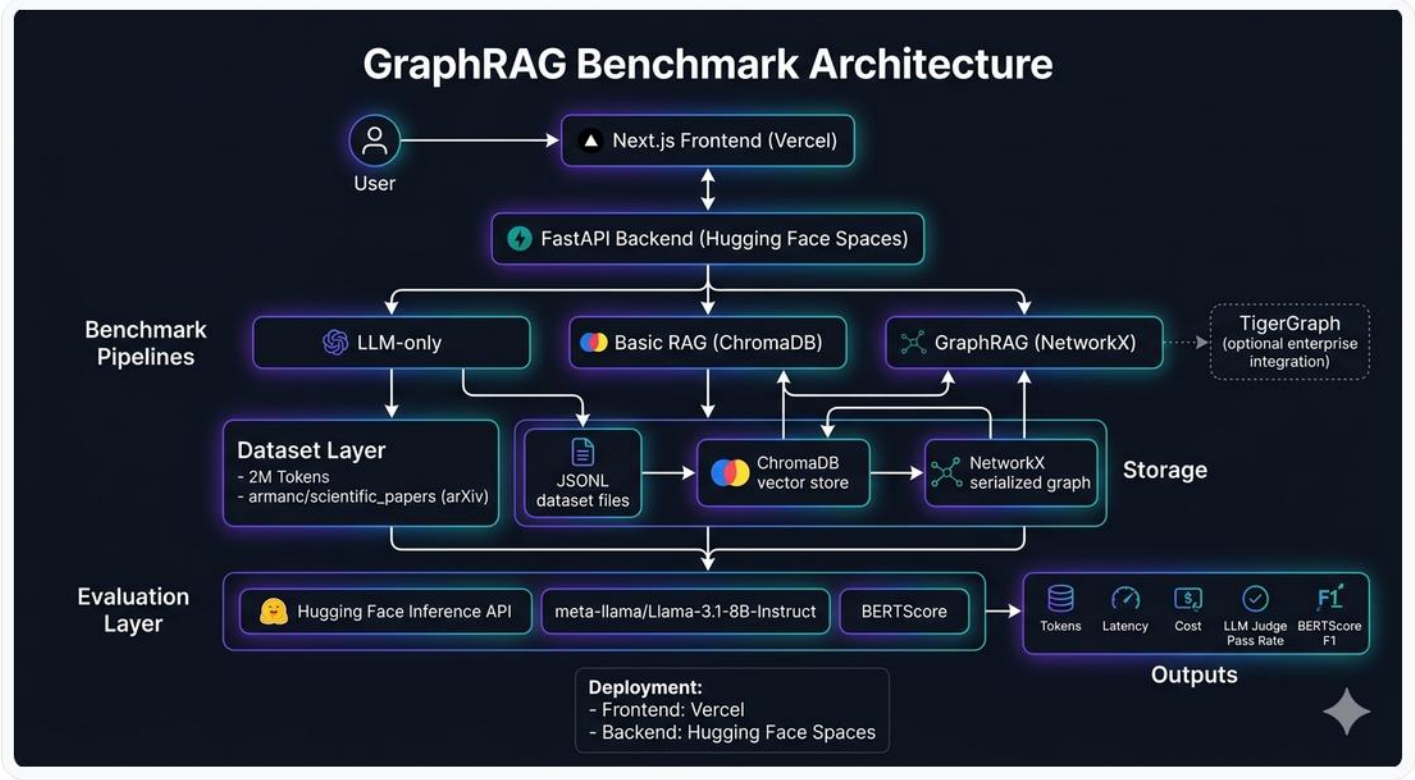
Vercel frontend + Hugging Face Spaces backend

Problem statement

LLM-only systems can hallucinate and lack source grounding. Basic RAG improves grounding with vector search but often misses relationships across entities, methods, datasets, and documents. This benchmark tests whether graph-structured retrieval can provide compact, relationship-aware context while preserving or improving answer quality.

Key conclusion

GraphRAG produced the strongest BERTScore in the current run and the lowest average latency, while using substantially fewer tokens than Basic RAG. The benchmark demonstrates the value of graph-based retrieval methodology without requiring TigerGraph.



The deployed architecture uses a Next.js frontend on Vercel and a FastAPI backend on Hugging Face Spaces. ChromaDB stores vector embeddings, NetworkX stores the serialized graph, and TigerGraph remains an optional enterprise connector.

Pipeline Methodology

LLM-only

Direct question to the language model. Useful baseline, but no retrieval grounding.

Basic RAG

ChromaDB vector search retrieves semantically similar chunks and sends them as context.

GraphRAG

NetworkX graph traversal retrieves relationship-aware chunks through entity and graph structure.

TigerGraph to NetworkX pivot

TigerGraph was explored first because it is powerful for enterprise graph storage. In practice, authentication issues, Docker complexity, resource requirements, and infrastructure overhead distracted from the research question. NetworkX preserves graph traversal and multi-hop reasoning, runs entirely in Python, and keeps the benchmark reproducible for judges.

Benchmark Results

Pipeline	Tokens	Latency	Cost	Judge	BERT F1
LLM-only	367.9	4.99s	\$0.00074	Not available	-0.080
Basic RAG	746.8	2.47s	\$0.00149	Not available	0.088
GraphRAG	415.0	1.75s	\$0.00083	Not available	0.110

LLM-as-a-Judge pass rate is marked unavailable because the current accuracy report contains null pass-rate values. This usually means hosted judge calls were skipped or unavailable during the evaluation run. BERTScore values are included from scientific_accuracy_report.json.

Vs LLM-only	Token	Latency	Cost
Basic RAG	-103.0%	50.5%	-103.0%
GraphRAG	-12.8%	65.0%	-12.8%

Accuracy and Conclusions

Accuracy methodology

Two complementary evaluations are used. LLM-as-a-Judge asks an independent hosted model to grade PASS or FAIL against a reference answer. BERTScore measures semantic similarity, capturing correct paraphrases that do not match exact wording.

Dataset rationale

Scientific papers are ideal for GraphRAG because they contain methods, datasets, tasks, concepts, citations, and cross-document relationships. The arXiv corpus creates realistic opportunities for multi-hop retrieval.

Key conclusions

- GraphRAG achieved the highest BERTScore F1 in the current accuracy report.
- GraphRAG had the lowest average latency among the three pipelines.
- GraphRAG used far fewer tokens and lower estimated cost than Basic RAG.
- NetworkX is sufficient for the benchmark because the claim is about graph-based retrieval methodology, not vendor-specific graph infrastructure.
- TigerGraph remains valuable as an optional enterprise connector for larger deployments.